

Workshop Sheet - Week 1

Reading and writing a Lean proof. InfoView displays your hypotheses and current goal. Use this tactic reference alongside the worked examples and syntax hints in the Lean file.

intro h	Assume the premise of an implication.
exact h	Submit h as a proof of the current goal.
apply h	Apply the result h.
rw [h]	Rewrite using an equality.
rfl	Prove an equality whose two sides are equal by definition.
use k	Choose k as a witness for an existence goal.
cases h	Split a hypothesis into its possible cases.
rcases h with ...	Recursive cases: unpack nested hypotheses using a pattern.
have h : P := by ...	State and prove an intermediate claim.
by_contra h	Assume the goal is false (call this h), to derive a contradiction.
simp	Simplify the goal using standard rewrite rules.
norm_num	Normalize numbers: simplify and prove numerical facts.
positivity	Prove that an expression is positive, nonnegative or nonzero.
linarith	Combine linear equations and inequalities ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$).
nlinarith	Deduce consequences of polynomial equalities and inequalities.

Indentation groups proof steps; `·` starts a separate proof branch.

1. Tactic practice.

- (a) Prove that any proposition P implies itself.
- (b) Given $k \in \mathbb{N}$, prove that there exists $m \in \mathbb{N}$ with $m = k$.
- (c) For integers a, b , deduce $a + 1 = b + 1$ from $a = b$.
- (d) For $m \in \mathbb{N}$, prove $(m + 1) - m = 1$.
- (e) For $n \in \mathbb{N}$, if $n = 2$ or $n = 3$, prove $n^2 \leq 9$.
- (f) For $x, y \in \mathbb{R}$ with $x + y \leq 7$ and $3 \leq y$, prove $x \leq 4$.
- (g) For $x \in \mathbb{R}$, prove $x^2 + 1 > 0$.
- (h) For $x \in \mathbb{R}$ with $x^2 \leq 4$, prove $x \leq 2$.
- (i) Verify that $\frac{3}{4} + \frac{1}{6} = \frac{11}{12}$ in $\mathbb{Q}$.

2. Infinitely many primes.

Parts (a)–(c) practise witnesses, also used in Exercise 3; (d)–(e) build Euclid's argument.

- (a) For each $n \in \mathbb{N}$, prove that there exists $m \in \mathbb{N}$ with $n < m$.
- (b) For $d, a, b \in \mathbb{N}$, prove that $d \mid a$ and $d \mid b$ imply $d \mid a + b$.
- (c) For $a, b, c \in \mathbb{N}$, prove that $a \mid b$ and $b \mid c$ imply $a \mid c$.
- (d) Prove that $n! + 1$ has a prime divisor.

Finding facts. Find factorial positivity and prime-divisor existence in [Loogle](#). Inspect the results with `#check`.

- (e) Suppose p is prime, $p \leq n$ and $p \mid n! + 1$. Derive a contradiction.
Hint. Search for divisibility of a factorial and divisibility of a difference. Show that p divides 1, contradicting primality.
- (f) Formalize the statement and proof of the following theorem.
Theorem. *There are infinitely many prime numbers.*

3. The irrationality of $\sqrt{2}$.

A rational square root would give coprime integers a, b with $a^2 = 2b^2$. In (f), combine (a)–(e) to rule this out. The Lean file supplies the number-system conversions.

- (a) For integers a, b satisfying $a^2 = 2b^2$, exhibit a witness proving $2 \mid a^2$.
- (b) For $a \in \mathbb{Z}$, deduce $2 \mid a$ from $2 \mid a^2$.
Hint. Search for a result about a prime dividing a product, then consider the two cases.
- (c) For integers a, b satisfying $a^2 = 2b^2$, prove that a is even.
- (d) Substitute $a = 2k$ into $a^2 = 2b^2$ and deduce $b^2 = 2k^2$.
Hint. Rewrite the hypothesis, then use `nlinarith`.
- (e) Show that coprime natural numbers a, b cannot both be even.
Hint. A common divisor divides the greatest common divisor.
- (f) Formalize the statement and proof of the following theorem.
Theorem. *The number $\sqrt{2}$ is irrational.*